Exploring the Roles and Effectiveness of Generative AI in Supporting CS Educators: A Systematic Literature Review

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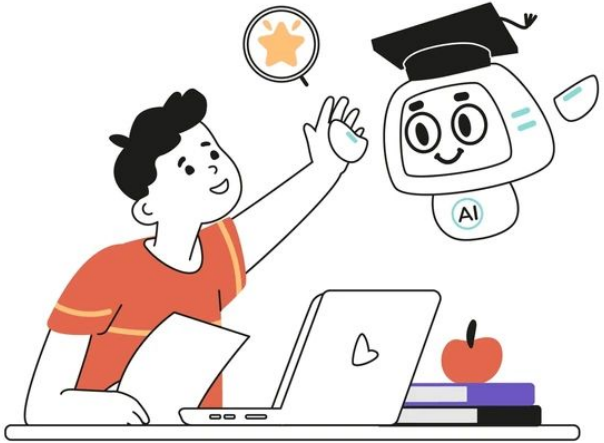
Presentation by : Cong Guo
Advisor : Dr. Amy Cook

Motivation: GenAI play multiple roles to support students



Tutor

Writing Coach

Study Buddy

Career Advisor

Motivation: Students reliance on GenAI causes concerns

Limited knowledge base

Critical thinking

Problem-solving abilities

Motivation: Leveraging GenAI to Enhance Educator Efficiency

Teachers possess rich subject knowledge

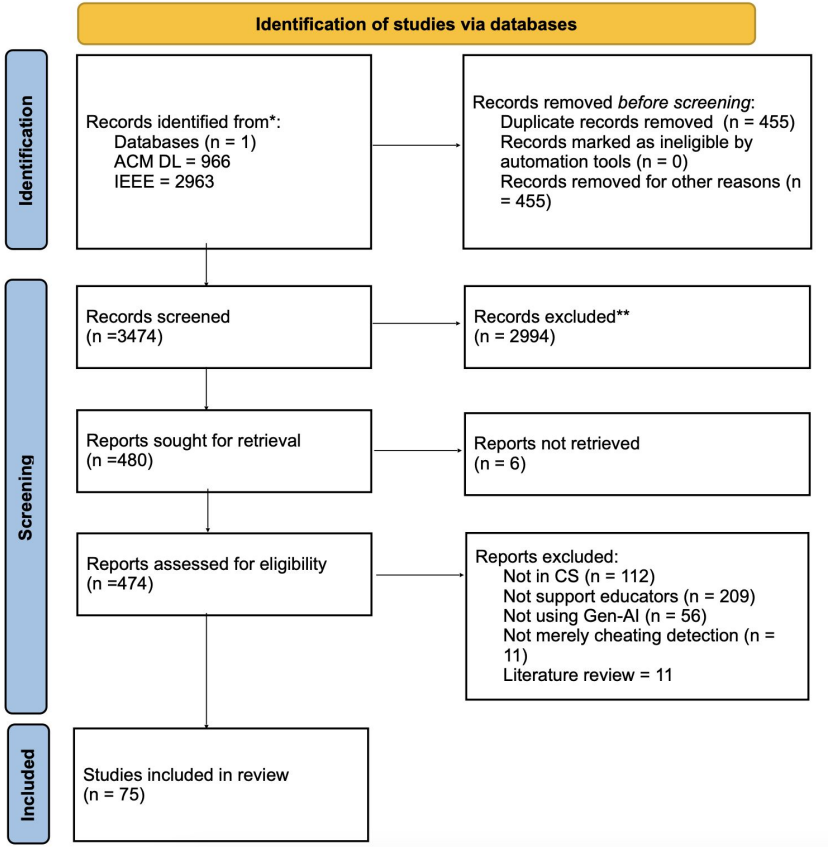
Research Questions

RQ1: What **role does GenAI play** in supporting computer science educators?

RQ2: How effectively does GenAI **perform** in educational contexts?

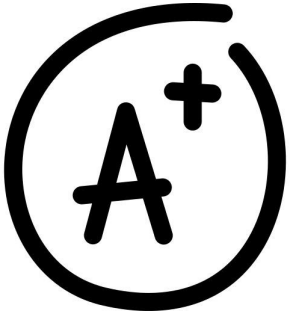
RQ3: What are directions for **future exploration** of GenAI in education?

Method: Systematic Literature Review

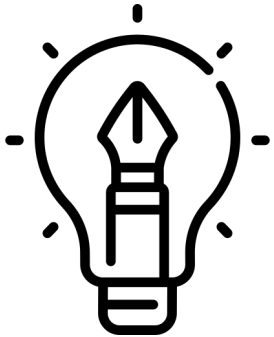


Findings: GenAI play Four primary roles to support educators

The Examiner



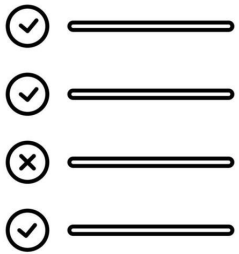
The Grader



The Designer

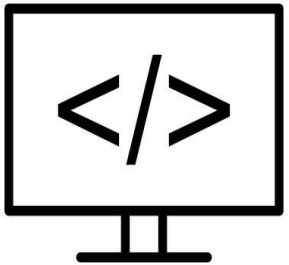
The Mentor

Role 1 Examiner: AI-generated exercises or quizzes (38)



Multiple-Choice Questions

[1,14,16,27,36,47,71]



Coding exercise

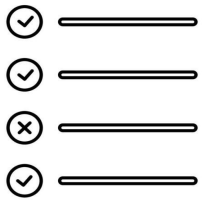
[2,3,4,5,12,13,15,17,18,21,25,30,32,33,35,39,40,42,46,56,60,62,63,66,69]

Open-ended Questions and Free Response Exercises

[22,44,51,54,55,68]

Role 1 Examiner: Strengths in GenAI Exercise Generation

1. AI effectively generates **high-quality** Exercises and Questions.(32 papers)
2. Using GenAI significant **reduction** in Educator **workload.** [1,2,4,12,15,18,33,39,42,46,54,55,68]
3. **Human-AI Hybrid** produce the best educational outcomes.



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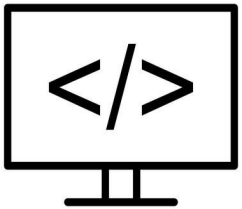

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Open-ended Questions and Free Response Exercises

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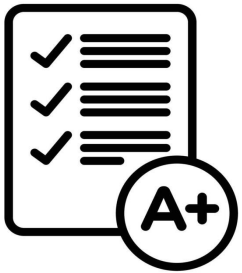
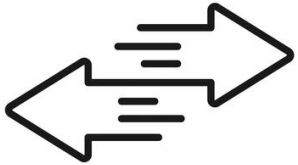
Role 2 Grader: AI-assisted grading and feedback (10)



Summative Grading

[19, 37, 50]



Both

[6,26,28,72]

Formative Feedback

[8,41,73]

Role 2: GenAI effectively assist automated grading and feedback

1. Automated grading and feedback accuracy **over 90%.**[8,72,50]
2. Models such as GPT-4 demonstrate performance **comparable to or exceeding** human evaluators.[6,19,26,28,37,41,72]

Summative Grading

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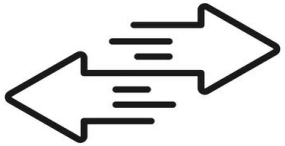

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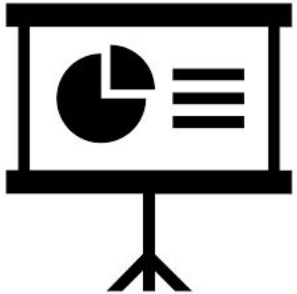
Role 3 Designer : AI-generated Pedagogical Materials (14)



Before Class:
Teaching Materials

[7,31,38,61,67]

During Class:
Interactivity Monitoring

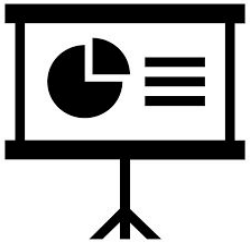
[11,24,34,70,74]

Across Courses:
Curricular Alignment

[9,29,52,57]

Role 3 Designer : Precision Challenges in AI-Generated Materials

1. AI outputs can be **misleading, or misaligned** with learning goals.
2. **Integrating educator feedback** effectively into AI outputs remains challenging.



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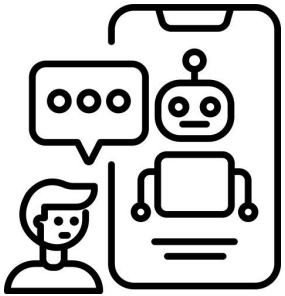
Role 4 Mentor: AI-Supported Mentorship in CS Education (13)



Virtual TAs or tutors

[10,20,23,45,65]

Teacher training

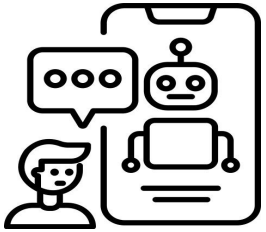
[49,64]

Instant course feedback

[43,48,53,58,59,75]

Role 4 Mentor: GenAI have limitations in Contextual Understand

1. AI's **accuracy is unstable** when dealing with complex and non-standard problems.
2. Reliance on AI-generated instant feedback **requires educators to analyze results and adjust instructional plans manually,** potentially increasing educators' workload.



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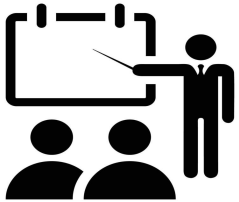

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Instant course feedback

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Implications: Existing limitations affect GenAi support educators

1. It is difficult to explore the **latest versions** of GenAI tools for support.
2. Most AI tools are designed for specific programming languages (e.g., Python, Java, SQL), which **limits** their usefulness **across a broader range of courses**.
3. Although AI tools help reduce educators' workload, assessing the **direct impact** on instructional quality remains challenging.

My Next Step: Interview educators to understand real needs

1. Interview educators to identify real-world **teaching challenges.**
2. Clarify **expectations** for AI tools in specific classroom scenarios and workload management.
3. Capture educators' **desired features**, **usability needs**, and current integration **barriers**.

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Thank You!